Alvee Ishraque

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Education

BRAC University
Bachelor of Science in Computer Science & Engineering (CSE) – CGPA: 3.64 / 4.00 2022 – January 2026

Research Interests

- Multi-Modal Machine Learning
- Deep Learning Architectures
- Medical Imaging
- Bioinformatics
- AI in Healthcare
- Federated Learning
- Computational Social Science
- AI/ML
- Natural Language Processing

Research Experience

Undergraduate Researcher | XEB-FuseNet: A Hierarchical Attention-Based Multimodal Fusion Framework

- Developed a multimodal deep learning framework fusing chest X-rays, ECGs, and blood data via hierarchical attention for thoracic disease diagnosis.
- Evaluated multiple backbones (e.g., Inception V3, DenseNet-121), achieving a peak AUC of 0.9052 and outperforming unimodal baselines.

Projects

AI-ML

Comparison of Word Representations for Multi-Class Text Classification

- Compared word representations (BoW, TF-IDF, GloVe, Skip-gram) with ML models (Logistic Regression, Naive Bayes, Random Forest) and NN models (DNN, RNN, GRU, Bidirectional LSTM) for multi-class text classification.
- Used 340,000 question-answer pairs across ten categories.

Flood Prediction Using Machine Learning

- Analyzed meteorological data, river flow, and land use patterns.
- Implemented decision trees, random forests, SVM, KNN, and neural networks to predict flood risk.

WEB

PetSphere: Full-Stack Pet Care Platform

- Built a MERN-stack app for pet adoption, healthcare, and services.
- Implemented REST APIs, JWT authentication, and role-based dashboards.
- Added pet profiles, medical logs, adoption workflows, and lost-and-found notifications.

Related Courses and Skills

Relevant Courses: CSE427: Machine Learning, CSE440: Natural Language Processing
Languages: Python, C, C++, JavaScript, SQL, Assembly
Web: MERN, HTML, PHP
Tools: Git, Google Workspace, Canva, MS Office Suite
English Proficiency: IELTS Academic – Overall 6.5

Extracurricular Activities

Senior Executive, Finance Department 2025 – 2026
BRAC University Environment Club

- Oversaw recruitment, onboarding, and team coordination for new club members.

Volunteer, 16th Convocation 2025
BRAC University

- Supported event operations, guest coordination, and logistics for the successful execution.

References

Dr. Amitabha Chakrabarty
Professor, Dept. of CSE, BRAC University
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Faculty Profile
Thesis Supervisor

Md. Saiful Bari Siddiqui
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Thesis Co-Supervisor